21-242 Recitation 12

Fields and Vector Spaces — Q4. Cancellation / + = implies =

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Show that if $\vec{v} + \vec{w} = \vec{v}$, then $\vec{w} = \vec{0}$.

Course-level solution

Add $-\vec{v}$ to both sides:

$$-\vec{v} + (\vec{v} + \vec{w}) = -\vec{v} + \vec{v} \implies (-\vec{v} + \vec{v}) + \vec{w} = \vec{0} \implies \vec{0} + \vec{w} = \vec{0} \implies \vec{w} = \vec{0}.$$

Later-course solution (using ideas from the rest of the handout)

This is the cancellation law in the abelian group $(V, +)$. Equivalently, the translation map $T_{\vec{v}}(\vec{w}) = \vec{v} + \vec{w}$ is injective (in fact invertible, with inverse $T_{-\vec{v}}$).